



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 20/07/2017*

EXERCISE

We observe N samples of a realization of the random process $X[n] = A \cdot s[n] + W[n]$, for $n=0, 1, \dots, N-1$, where A is the amplitude of the signal of interest, modelled as an unknown deterministic parameter, $s[n] = (-1)^n$ and $\{W[n]; n = 0, 1, \dots, N-1\}$ are independent and identically distributed (IID) random variables modeling additive noise. Each noise sample is Gaussian distributed, zero mean and with variance σ_w^2 .

- (1) Derive the maximum likelihood (ML) estimator of the unknown signal A when the noise power σ_w^2 is a priori known.
- (2) Derive the bias, the mean square error (MSE) of the ML estimator of A , and the Cramér-Rao lower bound (CRB) on the estimation of A . Comment on the consistency and efficiency of the ML estimator. Plot the MSE and the CRB as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-2} when the noise power is $\sigma_w^2 = 2$.
- (3) Derive the ML estimator of the unknown noise power σ_w^2 when we know that $A=0$.
- (4) Derive the bias, the MSE of the ML estimator of σ_w^2 , and the CRB on the estimation of σ_w^2 . Comment on the consistency and efficiency of the ML estimator. Plot the MSE and the CRB as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-2} when the noise power is $\sigma_w^2 = 2$.



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Solution.

(1) Derive the maximum likelihood (ML) estimator of the unknown signal A when the noise power σ_w^2 is a priori known.

$$X[n] \in N(A \cdot (-1)^n, \sigma_w^2) \Rightarrow X \in N(A \cdot \mathbf{s}, \sigma_w^2 \mathbf{I})$$

$$\hat{A}_{ML} = \arg \max_A f_X(\mathbf{x}; A) = \arg \min_A \|\mathbf{x} - A \cdot \mathbf{s}\|^2 = \hat{A}_{LS} = \mathbf{s}^\# \mathbf{x}$$

where $\mathbf{s}^\# \triangleq (\mathbf{s}^H \mathbf{s})^{-1} \mathbf{s}^H = \frac{\mathbf{s}^T}{N}$ Moore-Penrose pseudoinverse

$$\hat{A}_{ML} = \frac{\mathbf{s}^T \mathbf{x}}{N} = \frac{1}{N} \sum_{n=0}^{N-1} (-1)^n X[n]$$

(2) Derive the bias, the mean square error (MSE) of the ML estimator of A , and the Cramér-Rao lower bound (CRB) on the estimation of A . Comment on the consistency and efficiency of the ML estimator. Plot the MSE and the CRB as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-2} when the noise power is $\sigma_w^2 = 2$.

$$E\{\hat{A}_{ML}\} = \frac{\mathbf{s}^T E\{\mathbf{x}\}}{N} = \frac{A}{N} \mathbf{s}^T \mathbf{s} = A \Rightarrow b\{\hat{A}_{ML}\} = E\{A - \hat{A}_{ML}\} = 0 \Rightarrow \text{unbiased estimator}$$

$$MSE\{\hat{A}_{ML}\} = \text{var}\{\hat{A}_{ML}\} = \frac{1}{N^2} \text{var}\left\{\sum_{n=0}^{N-1} (-1)^n X[n]\right\} = \frac{1}{N^2} \sum_{n=0}^{N-1} (-1)^{2n} \sigma_w^2 = \frac{\sigma_w^2}{N} \Rightarrow \text{consistent estimator}$$

since the data samples $X[n]$'s are mutually independent and they have the same variance σ_w^2 .

$$CRB(A) = \frac{\sigma_w^2}{N} = MSE\{\hat{A}_{ML}\} \Rightarrow \text{efficient estimator}$$

$$MSE\{\hat{A}_{ML}\} = \frac{\sigma_w^2}{N} < 10^{-2} \rightarrow N > 10^2 \sigma_w^2 = 200$$



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(3) Derive the ML estimator of the unknown noise power σ_w^2 when we know that $A=0$.

$$X[n] \in N(0, \sigma_w^2) \Rightarrow X \in N(\mathbf{0}, \sigma_w^2 \mathbf{I})$$

$$\hat{\sigma}_{w,ML}^2 = \arg \max_{\sigma_w^2} (\ln f_X(\mathbf{x}; \sigma_w^2)) = \arg \max_{\sigma_w^2} \left(-\frac{N}{2} \ln(2\pi\sigma_w^2) - \frac{\|\mathbf{x}\|^2}{2\sigma_w^2} \right)$$

Calculating the derivative of the log-LF with respect to σ_w^2 and setting the derivative equal to 0 we get:

$$\hat{\sigma}_{w,ML}^2 = \frac{1}{N} \|\mathbf{x}\|^2 = \frac{1}{N} \sum_{n=0}^{N-1} X^2[n]$$

(4) Derive the bias, the MSE of the ML estimator of σ_w^2 , and the CRB on the estimation of σ_w^2 . Comment on the consistency and efficiency of the ML estimator. Plot the MSE and the CRB as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-2} when the noise power is $\sigma_w^2 = 2$.

$$E\{\hat{\sigma}_{w,ML}^2\} = \frac{1}{N} \sum_{n=0}^{N-1} E\{X^2[n]\} = \sigma_w^2 \Rightarrow b\{\hat{\sigma}_{w,ML}^2\} = E\{\sigma_w^2 - \hat{\sigma}_{w,ML}^2\} = 0 \Rightarrow \text{unbiased estimator}$$

$$MSE\{\hat{\sigma}_{w,ML}^2\} = \text{var}\{\hat{\sigma}_{w,ML}^2\} = \frac{1}{N^2} \text{var}\left\{\sum_{n=0}^{N-1} X^2[n]\right\} = \frac{1}{N} \text{var}\{X^2[n]\} = \frac{2\sigma_w^4}{N} \Rightarrow \text{consistent estimator}$$

$$\text{var}\{X^2[n]\} = E\{X^4[n]\} - (E\{X^2[n]\})^2 = 3\sigma_w^4 - \sigma_w^4 = 2\sigma_w^4$$

since the data samples $X[n]$'s are Gaussian distributed, mutually independent and they have the same variance σ_w^2 .

$$CRB(\sigma_w^2) = \frac{2\sigma_w^4}{N} = MSE\{\hat{\sigma}_{w,ML}^2\} \Rightarrow \text{efficient estimator}$$

$$MSE\{\hat{\sigma}_{w,ML}^2\} = \frac{2\sigma_w^4}{N} < 10^{-2} \rightarrow N > 10^2 \cdot 2\sigma_w^4 = 800$$

Note that in order to achieve the same estimation accuracy, we need more samples in the case of the estimation of the variance than in the case of the estimation of the amplitude (i.e. the mean).